

# CREATING DATABASE

```
CREATE database NORTHWIND_TRADERS;
```

```
ALTER TABLE categories
```

```
ADD CONSTRAINT pk_categories PRIMARY KEY (categoryID);
```

```
ALTER TABLE customers
```

```
MODIFY customerID VARCHAR(50) NOT NULL;
```

```
ALTER TABLE customers
```

```
ADD CONSTRAINT pk_customers PRIMARY KEY (customerID);
```

```
ALTER TABLE employees
```

```
ADD CONSTRAINT pk_employees PRIMARY KEY (employeeID);
```

```
ALTER TABLE orders
```

```
ADD CONSTRAINT orders PRIMARY KEY (orderID);
```

```
ALTER TABLE products
```

```
ADD CONSTRAINT products PRIMARY KEY (productID);
```

```
ALTER TABLE shippers
```

```
ADD CONSTRAINT pk_shippers PRIMARY KEY (shipperID);
```

```
ALTER TABLE customers
```

```
MODIFY customerID VARCHAR(50) NOT NULL;
```

ALTER TABLE orders

MODIFY customerID VARCHAR(50) NOT NULL;

ALTER TABLE orders

ADD CONSTRAINT fk\_orders

FOREIGN KEY (customerID) REFERENCES customers(customerID);

DELETE FROM employees

WHERE employeeID IS NULL;

DELETE FROM orders

WHERE employeeID IS NULL;

SELECT DISTINCT employeeID

FROM orders

WHERE employeeID NOT IN (SELECT employeeID FROM employees)

OR employeeID IS NULL;

INSERT INTO employees (employeeID, employeeName, title, city, country, reportsTo)

VALUES (2, 'Andrew Fuller', 'Vice President Sales', 'New York', 'USA', NULL);

ALTER TABLE orders

ADD CONSTRAINT fk\_orders\_employees

FOREIGN KEY (employeeID) REFERENCES employees(employeeID);

DELETE FROM shippers

WHERE shipperID IS NULL;

```
ALTER TABLE orders
```

```
ADD CONSTRAINT fk_orders_shippers
```

```
FOREIGN KEY (shipperID) REFERENCES shippers(shipperID);
```

```
ALTER TABLE order_details
```

```
ADD CONSTRAINT fk_orders_details_orders
```

```
FOREIGN KEY (orderID) REFERENCES orders(orderID);
```

```
DELETE FROM products
```

```
WHERE productID IS NULL;
```

```
DELETE FROM categories
```

```
WHERE categoryID IS NULL;
```

```
ALTER TABLE order_details
```

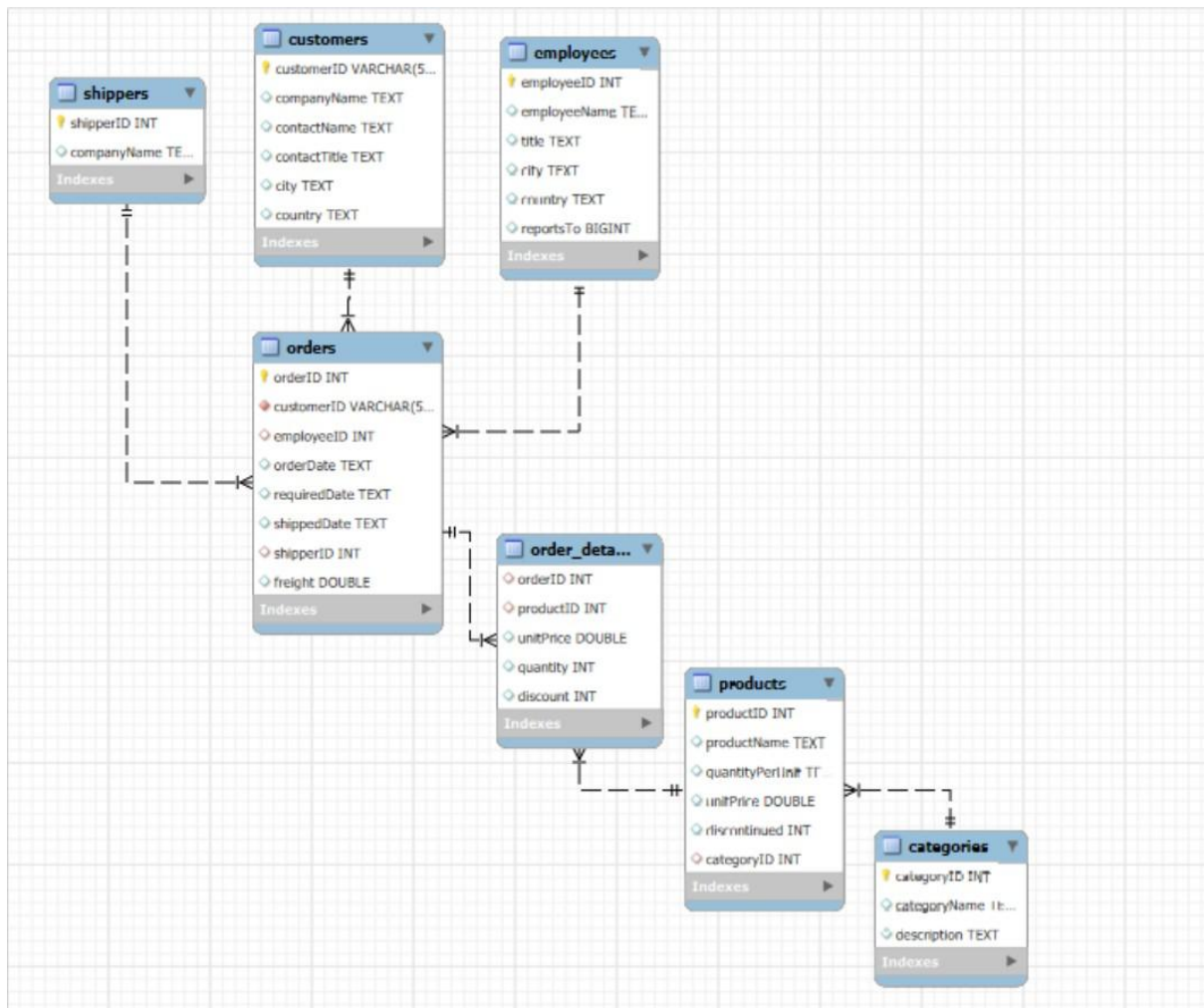
```
ADD CONSTRAINT fk_orders_details_products
```

```
FOREIGN KEY (productID) REFERENCES products(productID);
```

```
ALTER TABLE products
```

```
ADD CONSTRAINT fk_products_categories
```

```
FOREIGN KEY (categoryID) REFERENCES categories(categoryID);
```



-- SALES THROUGHT TIME

CREATE VIEW sales\_throuht\_time AS (

SELECT

YEAR(o.orderDate) as Year\_,

MONTH(o.orderDate) as Month\_,

SUM(od.quantity) as TotalQuantity,

SUM(od.quantity \* od.unitPrice) as TotalRevenue,

SUM(o.freight) as TotalFreightCosts,

SUM(od.quantity \* od.unitPrice) - SUM(o.freight) as TotalNetRevenue

FROM orders o

```
LEFT JOIN order_details od ON o.orderID = od.orderID

GROUP BY

    YEAR(o.orderDate),

    MONTH(o.orderDate)

);
```

-- CATEGORIES OF PRODUCTS THROUGHT TIME

```
CREATE VIEW product_categories_throuht_time AS (

    SELECT

        c.categoryName,

        YEAR(o.orderDate) as Year_,

        MONTH(o.orderDate) as Month_,

        SUM(od.quantity) as TotalQuantity,

        SUM(od.quantity * od.unitPrice) as TotalRevenue,

        SUM(o.freight) as TotalFreightCosts,

        SUM(od.quantity * od.unitPrice) - SUM(o.freight) as TotalNetRevenue

    FROM orders o

        LEFT JOIN order_details od ON o.orderID = od.orderID

        LEFT JOIN products p ON od.productID = p.productID

        LEFT JOIN categories c ON p.categoryID = c.categoryID

    GROUP BY

        c.categoryName,

        YEAR(o.orderDate),

        MONTH(o.orderDate)

);
```

-- PRODUCTS THROUGHT TIME

```

CREATE VIEW products_throuht_time AS (
    SELECT
        p.productName,
        YEAR(o.orderDate) as Year_,
        MONTH(o.orderDate) as Month_,
        SUM(od.quantity) as TotalQuantity,
        SUM(od.quantity * od.unitPrice) as TotalRevenue,
        SUM(o.freight) as TotalFreightCosts,
        SUM(od.quantity * od.unitPrice) - SUM(o.freight) as TotalNetRevenue
    FROM orders o
    LEFT JOIN order_details od ON o.orderID = od.orderID
    LEFT JOIN products p ON od.productID = p.productID
    LEFT JOIN categories c ON p.categoryID = c.categoryID
    GROUP BY
        p.productName,
        YEAR(o.orderDate),
        MONTH(o.orderDate)
);

```

-- CUSTOMERS THROUGHT TIME

```

CREATE VIEW customers_throught_time AS (
    SELECT
        c.companyName,
        c.country,
        c.city,
        YEAR(o.orderDate) AS Year_,

```

```

MONTH(o.orderDate) AS Month_,
SUM(od.quantity) AS TotalQuantity,
SUM(od.quantity * od.unitPrice) AS TotalRevenue,
SUM(o.freight) AS TotalFreightCosts,
SUM(od.quantity * od.unitPrice) - SUM(o.freight) AS TotalNetRevenue
FROM orders o
LEFT JOIN order_details od ON o.orderID = od.orderID
LEFT JOIN customers c ON o.customerID = c.customerID
GROUP BY
    c.companyName,
    c.country,
    c.city,
    YEAR(o.orderDate),
    MONTH(o.orderDate)
);

```

-- SHIPPERS TO CUSTOMERS QUANTITY AND REVENUE THROUGHT TIME

```

CREATE VIEW shippers_throught_time_per_country AS (
    SELECT
        c.country,
        YEAR(o.orderDate) AS Year_,
        MONTH(o.orderDate) AS Month_,

        -- SPEEDY EXPRESS
        SUM(CASE WHEN s.companyName = 'Speedy Express' THEN od.quantity
ELSE 0 END) AS Total_Speedy_Express_Quantity,

```

SUM(CASE WHEN s.companyName = 'Speedy Express' THEN od.quantity  
\* od.unitPrice ELSE 0 END) AS Total\_Speedy\_Express\_Revenue,

SUM(CASE WHEN s.companyName = 'Speedy Express' THEN o.freight  
ELSE 0 END) AS Total\_Speedy\_Express\_FreightCosts,

SUM(CASE WHEN s.companyName = 'Speedy Express' THEN od.quantity  
\* od.unitPrice ELSE 0 END)

- SUM(CASE WHEN s.companyName = 'Speedy Express' THEN o.freight  
ELSE 0 END)

AS Total\_Speedy\_Express\_NetRevenue,

-- UNITED PACKAGE

SUM(CASE WHEN s.companyName = 'United Package' THEN od.quantity  
ELSE 0 END) AS Total\_United\_Package\_Quantity,

SUM(CASE WHEN s.companyName = 'United Package' THEN od.quantity  
\* od.unitPrice ELSE 0 END) AS Total\_United\_Package\_Revenue,

SUM(CASE WHEN s.companyName = 'United Package' THEN o.freight  
ELSE 0 END) AS Total\_United\_Package\_FreightCosts,

SUM(CASE WHEN s.companyName = 'United Package' THEN od.quantity  
\* od.unitPrice ELSE 0 END)

- SUM(CASE WHEN s.companyName = 'United Package' THEN o.freight  
ELSE 0 END)

AS Total\_United\_Package\_NetRevenue,

-- FEDERAL SHIPPING

SUM(CASE WHEN s.companyName = 'Federal Shipping' THEN od.quantity  
ELSE 0 END) AS Total\_Federal\_Shipping\_Quantity,

SUM(CASE WHEN s.companyName = 'Federal Shipping' THEN od.quantity  
\* od.unitPrice ELSE 0 END) AS Total\_Federal\_Shipping\_Revenue,

SUM(CASE WHEN s.companyName = 'Federal Shipping' THEN o.freight  
ELSE 0 END) AS Total\_Federal\_Shipping\_FreightCosts,

SUM(CASE WHEN s.companyName = 'Federal Shipping' THEN od.quantity  
\* od.unitPrice ELSE 0 END)



- SUM(CASE WHEN s.companyName = 'Federal Shipping' THEN o.freight  
ELSE 0 END)

AS Total\_Federal\_Shipping\_NetRevenue

FROM order\_details od

LEFT JOIN orders o ON od.orderID = o.orderID

LEFT JOIN customers c ON o.customerID = c.customerID

LEFT JOIN shippers s ON o.shipperID = s.shipperID

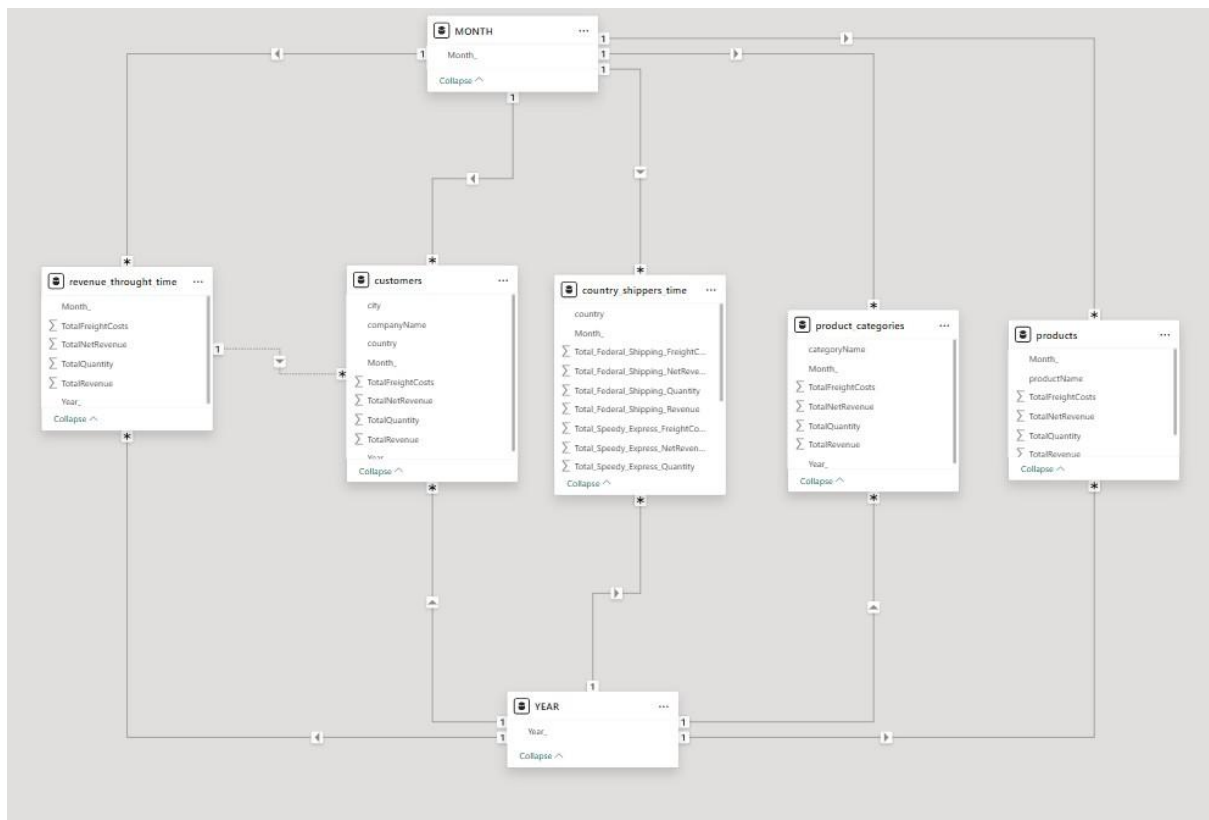
GROUP BY

c.country,

YEAR(o.orderDate),

MONTH(o.orderDate)

);



| TOTAL QUANTITY | TOTAL REVENUE | TOTAL FREIGHT COSTS | TOTAL NETO REVENUE | Value per product | Net Value per product |
|----------------|---------------|---------------------|--------------------|-------------------|-----------------------|
| 51K            | 1,35M         | 207,31K             | 1,15M              | 26,39             | 22,35                 |

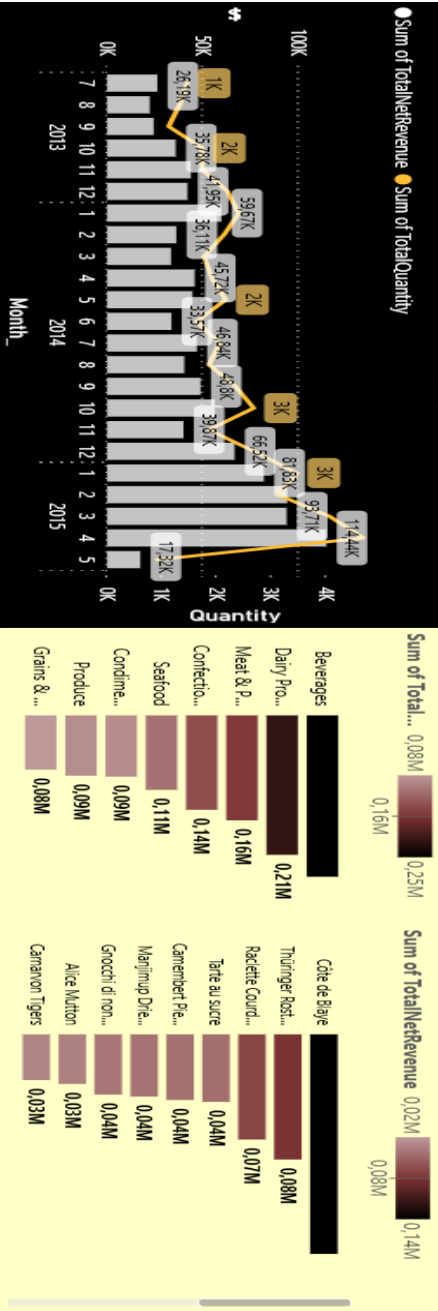
Year\_

201320142015

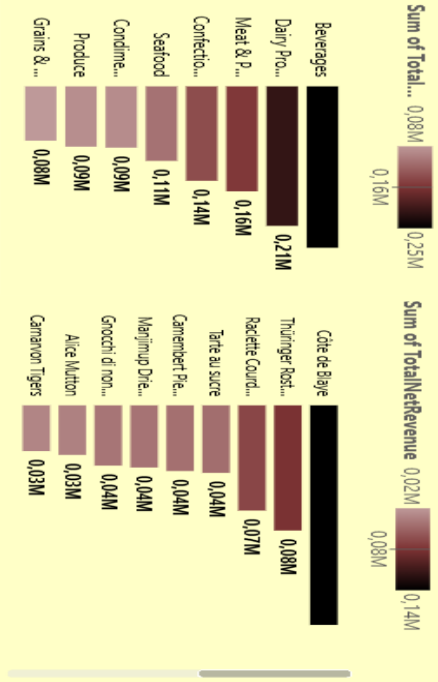
Month\_

123456789101112

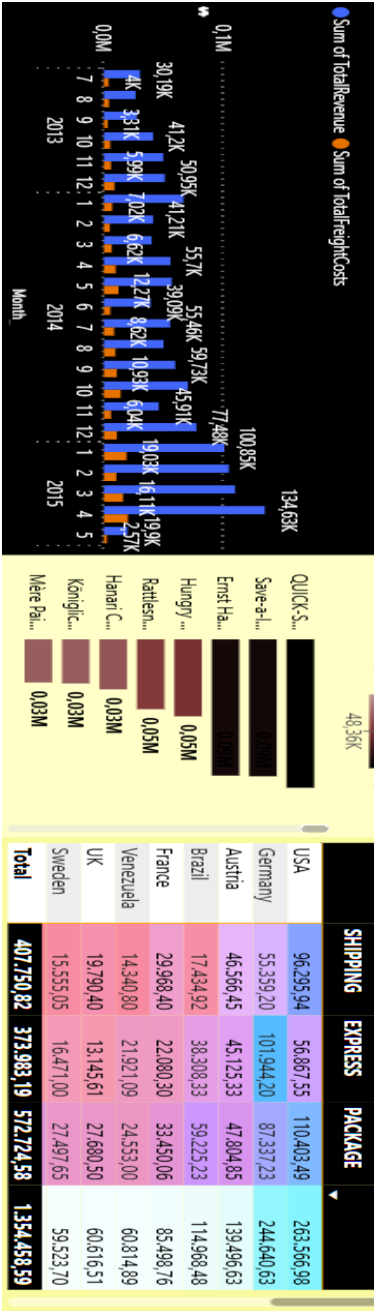
Sum of TotalNetRevenue and Sum of TotalQuantity by Year\_ and Month\_



TOTAL NET REVENUE BY CATEGORIES



Sum of TotalRevenue and Sum of TotalFreightCosts by Year\_ and Month\_



| country   | REVENUE<br>FEDERAL<br>SHIPPING | REVENUE<br>SPEED<br>EXPRESS | REVENUE<br>UNITED<br>PACKAGE | REVENUE<br>TOTAL |
|-----------|--------------------------------|-----------------------------|------------------------------|------------------|
| USA       | 96.295.94                      | 56.067.55                   | 110.403.49                   | 263.566.98       |
| Germany   | 55.359.20                      | 101.944.20                  | 87.337.23                    | 244.640.63       |
| Austria   | 46.566.45                      | 45.125.33                   | 47.804.85                    | 139.496.63       |
| Brazil    | 17.434.92                      | 38.308.33                   | 59.225.23                    | 114.968.48       |
| France    | 29.968.40                      | 22.080.30                   | 33.450.06                    | 85.498.76        |
| Venezuela | 14.340.80                      | 21.921.09                   | 24.553.00                    | 60.814.89        |
| UK        | 19.790.40                      | 13.145.61                   | 27.680.50                    | 60.616.51        |
| Sweden    | 15.555.05                      | 16.471.00                   | 27.497.65                    | 59.523.70        |
| Total     | 407.750.82                     | 373.983.19                  | 572.724.58                   | 1.354.458.59     |